



Total saponins from *Panax japonicus* reduced lipid deposition and inflammation in hepatocyte via PHD2 and hepatic macrophage-derived exosomal miR-463-5p

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ABSTRACT

Ethnopharmacological relevance: *Panax japonicus* (T. Nees) C.A. Mey. (PJ) is a traditional Chinese herbal medicine revered as the “King of Herbs” in Tujia and Hmong medical practices. Clinically, it is primarily used to treat weakness and fatigue, wound bleeding, arthritis, hyperlipidemia, and fatty liver. It is rich in saponins, and the total saponins from PJ (TSPJ), possess immunomodulatory, antioxidant, and lipid-lowering effects. These properties hold significant potential in managing liver-related metabolic diseases such as non-alcoholic fatty liver disease (NAFLD) and non-alcoholic steatohepatitis (NASH).

Aim of study: Evaluate the therapeutic effects of TSPJ on lipid metabolism disorders in a NASH model and explore the possible underlying mechanisms.

Materials and methods: To model NASH, C57BL/6J mice were fed a high-fat diet (HFD) and RAW264.7 cells were stimulated with lipopolysaccharide (LPS) and palmitic acid (PA). The animal and cell models were also treated with TSPJ, and the changes in inflammation and lipid metabolism were measured. Additional models were created by transfecting lentiviral vectors to cause miR-463-5p knockdown in the C57BL/6J mouse and the RAW264.7 cells.

Results: In the HFD-induced mice, TSPJ reduced the body weight and liver weight, lowered the serum levels of TG, T-CHO, ALT, and AST, and reduced the hepatic lipid droplet formation and vacuolization. In the RAW264.7 cells, TSPJ upregulated the M2 markers and downregulated the M1 markers. TSPJ also significantly increased the expression of miR-463-5p in the exosomes derived from the RAW264.7 cells or the primary mouse hepatic macrophages, and miR-463-5p suppressed the expression of PHD2 in hepatocytes to improve lipid metabolism. However, when the exosome secretion inhibitor GW4869 was applied, TSPJ became less effective in alleviating the lipid deposition and inflammation in hepatocytes.

Conclusions: TSPJ significantly upregulated the expression of miR-463-5p in the exosomes of hepatic macrophages to thus downregulate PHD2 expression in hepatocytes and improve hepatic lipid metabolism.

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Abbreviations:

ACC1	Acetyl-CoA carboxylase 1
ALT	Alanine transaminase
ANGPTL4	Angiopoietin-like 4
AST	Aspartate aminotransferase
BSA	Bovine serum albumin
CD9	Cluster of Differentiation 9
cDNA	Complementary deoxyribonucleic acid
DAB	3,3'-Diaminobenzidine
DAPI	4',6-Diamidino-2-phenylindole
DMEM	Dulbecco's Modified Eagle Medium
DMSO	Dimethyl sulfoxide
DSS	Dextran Sulfate Sodium Salt
ECL	Enhanced chemiluminescence
Exos.	Exosomes
FASN	Fatty acid synthase
FBS	Fetal bovine serum
GBSS	Gey's balanced salt solution
HDL-C	High-density lipoprotein cholesterol
HIF-1α	Hypoxia-inducible factor 1-α
HPLC	High performance liquid chromatography
HSP70	Heat shock protein 70
IL-10	Interleukin-10
IL-1α	Interleukin-1α
IL-6	Interleukin-6
LV	Lentivirus
miRNA	MicroRNA
mRNA	Messenger RNA
PBS	Phosphate buffered saline
PG	<i>Panax ginseng</i> C.A. Mey.
PHD2	Prolyl Hydroxylase Domain 2
PJ	<i>Panax japonicus</i> (T. Nees) C.A. Mey.
PPAR-α	Peroxisome proliferator-activated receptor alpha
PVDF	Polyvinylidene fluoride
pVHL	von Hippel-Lindau protein
rpm	Revolutions per minute
SDS-PAGE	Sodium dodecyl sulfate polyacrylamide gel electrophoresis
SREBP-1C	Sterol Regulatory Element Binding Protein-1C
Sup.	Supernatant
TBST	Tris-Buffered Saline with Tween
T-CHO	Total cholesterol
TG	Triglycerides
TNFα	Tumor necrosis factor α
TSG101	Tumor susceptibility gene 101 protein
TSPJ	Total saponins from <i>Panax japonicus</i> (T. Nees) C.A. Mey.

1. Introduction

Non-alcoholic fatty liver disease (NAFLD) has emerged as a significant global public health issue, affecting approximately 25% of the world's population (Guo et al., 2022; Kaya and Yilmaz, 2020; Wang et al., 2023). It is the most common chronic liver disease, and its prevalence in Western countries is up to 35% (Kounatidis et al., 2024). The early stage of NAFLD is characterized by the substantial accumulation of fat in the liver. Without effective treatment, it can lead to non-alcoholic steatohepatitis (NASH), cirrhosis, and even liver cancer (Koh et al., 2024). However, effective clinical prevention and treatment measures are currently lacking. Management focuses on lifestyle modifications, including weight loss, diet, and exercise.

As a severe form of NAFLD, NASH is characterized by liver inflammation and damage, in addition to fat accumulation in the liver (European Association for the Study of the Liver, the European Association for the Study of Diabetes, and the European Association for the Study of Obesity). It is a progressive condition that can lead to serious liver complications. It affects approximately 1.5%–6.5% of the adults in the U.S., and its prevalence is expected to rise significantly. Its pathophysiology involves hepatocyte injury, inflammation, and fibrosis, and it can progress to advanced liver fibrosis, cirrhosis, and hepatocellular carcinoma. It is a leading cause of liver-related morbidity and mortality, and liver transplantation may be necessary in severe cases. The exact mechanisms are complex and involve insulin resistance, oxidative stress,

and inflammatory pathways. Pharmacological treatments are under investigation, with some showing promise in reducing liver inflammation and fibrosis.

Oxygen-sensitive pathways have garnered significant attention for their crucial role in metabolic disorders, and prolyl hydroxylase domain (PHD) proteins are key rate-limiting enzymes in lipid metabolism (Matsuura et al., 2013). PHD proteins exist in three isoforms, and PHD2 is a key regulatory molecule of hypoxia-inducible factor-1 α (HIF-1α) (Marsch et al., 2016; Song et al., 2022; Yun et al., 2002). Studies have shown that silencing HIF-1α inhibits PPAR-α/ANGPTL4 signaling and exacerbates cellular lipid accumulation (He et al., 2021). Moreover, HIF-1α can protect HepG2 cells from fatty acid-induced toxicity (Yoo et al., 2014). Thus, controlling HIF-1α expression represents a potential target for the treatment of lipid metabolism disorders, and the inhibition of PHD2 is an ideal strategy (Kaelin and Ratcliffe, 2008; Matsuura et al., 2013; Semenza et al., 1994; Taniguchi et al., 2013). It has been reported that the loss of PHD2 in adipocytes can mitigate the obesity induced by high-fat diet (HFD) and remedy lipid metabolism abnormalities (Li et al., 2023). The knockdown of PHD2 reduces the weight of liver and white adipose tissue as well as the adipocyte size in mice (Saito et al., 2019), and decreases the serum levels of T-CHO and HDL-C (Rahtu-Korpela et al., 2014, 2016). The PHD2 inhibitor enarodustat significantly reduces hepatic lipid droplet vacuolization and promotes the expression of HIF-1α and PPAR (Sugahara et al., 2020).

Exosomes are small vesicles (approximately 30–150 nm in diameter) capable of carrying various bioactive components, including proteins and nucleic acids, for intracellular information transmission (Kalluri and LeBleu, 2020). MicroRNAs (miRNAs) are a kind of tiny RNA molecules that do not encode proteins but control gene expression mostly post-transcriptionally by targeting mRNA, and they have been implicated as critical components of exosomes (Isaac et al., 2021; Krol et al., 2010). Exosome-mediated miRNA shuttling plays a crucial role in regulating metabolism-related genes (Yu et al., 2016), and several miRNAs are known to target and regulate the expression of PHD2 (Chen et al., 2016).

Resident macrophages, specifically Kupffer cells (KCs), are the predominant immune cells in the liver. It is noteworthy that obesity contributes to the activation of KCs, which leads to the recruitment of monocyte-derived macrophages into the liver, exacerbates obesity-induced liver inflammation, and alters the release of miRNAs via exosomes (Davis, 2007; Lackey and Olefsky, 2016). Ismail et al. found that macrophages transmit miR-223 via exosomes to various cells, including hepatocytes, thus affecting cellular functions (Ismail et al., 2013). Aucher et al. reported that human macrophages transfer miR-142 and miR-223 to liver cancer cells through exosomes, thereby inhibiting cancer cell proliferation (Aucher et al., 2013). Wu et al. reported that liver-derived exosomes can deliver miR-130a-3p to adipocytes to improve their glucose intolerance (Wu et al., 2020). Gao et al. discovered that KCs produce endogenous miR-690, which travels via exosomes to the neighboring hepatocytes and inhibits de novo lipogenesis, but the miR-690 levels are reduced in the hepatocytes of NASH patients (Gao et al., 2022). These findings reveal that macrophage-derived exosomes have a crucial impact on inflammation and lipid metabolism disorders.

Panax japonicus (T. Nees) C.A. Mey. (PJ) is a plant in the *Panax* genus of the Araliaceae family. It is widely distributed in East Asian regions such as China, Japan, and South Korea, and is often used as a substitute for *Panax ginseng* C.A. Mey. (PG) (Morita et al., 1985; Yang et al., 2014). It has been reported that PG is used in East Asia and Europe as a remedy for chronic diseases such as atherosclerosis, hyperlipidemia, hypertension, and non-insulin-dependent diabetes mellitus (Han et al., 1998; Inoue et al., 1999; Kiefer and Pantuso, 2003; Nocerino et al., 2000; Sotaniemi et al., 1995). In China, PJ is regarded as the “King of Herbs” in traditional Tujia and Hmong medicinal practices, thanks to its nourishing, hemostatic, anti-inflammatory, and liver-protective properties (Chen et al., 2023). Clinically, it is primarily used to treat weakness and

fatigue, wound bleeding, arthritis, hyperlipidemia, and fatty liver (Chen et al., 2023; Ngan et al., 1999; Zhang et al., 2015). This herb is rich in saponins, which are its main active components, and the total saponins from PJ (TSPJ) have immunoregulatory, antioxidative, lipid-lowering, and various other pharmacological properties (Chen et al., 2023; Yoshizaki et al., 2013). Research has found that the total Chikusetsu-saponins isolated from PJ have anti-obesity effects on HFD-fed mice by inhibiting pancreatic lipase activity (Han et al., 2005). Our previous studies have shown that TSPJ can mitigate lipid metabolism disorders in aging mice by regulating the intestinal flora (Hu et al., 2024). In this study, we created NASH models using C57BL/6J mouse, the RAW264.7 cell line, and the AML12 cell line to critically examine the crosstalk between hepatic macrophages and hepatocytes, and further elucidate the mechanisms by which TSPJ remedy hepatic lipid metabolism disorders. This work aimed to provide insights in understanding the pathogenesis of NAFLD and developing anti-NAFLD drugs for clinical practice.

2. Materials and methods

2.1. Extraction and preparation of TSPJ

PJ was collected from the Chunmuying Authentic Medicinal Herb Production Base (Enshi, Hubei, China) and identified by Dr. Ding Yuan at Key Laboratory of Natural Products Research and Utilization, China Three Gorges University. The name of the plant was further verified at the World Flora Online database (<https://www.worldfloraonline.org/>). The voucher specimen was deposited at the Institute of Chinese Herb Medicine, College of Medicine and Health Science, China Three Gorges University.

TSPJ was extracted and analyzed as previously described (He et al., 2008; Yuan et al., 2017). The dried roots of PJ were extracted with 60% ethanol at reflux (3 × 2 h). The combined liquid was concentrated to about 20% of the initial volume, then extracted sequentially with petroleum ether, ethyl acetate, and *n*-butanol. The *n*-butanol extract was loaded onto a column packed with D101 macroporous resin (Tianjin Pesticide Factory, Tianjin, China) that was then flushed with double-distilled water until the eluent was colorless. Finally, the column was washed with 60% ethanol, and the eluate was concentrated and freeze-dried to give TSPJ as a white powder in 18.4% yield (He et al., 2012). According to high performance liquid chromatography (HPLC), the saponin content of TSPJ was 83.48% (He et al., 2008; Hu et al., 2024; Shi et al., 2021).

2.2. Animal experiments

Eighty male C57BL/6J (8–20 g) mice were obtained from the Experimental Animal Center of China Three Gorges University (SCXK(E) 2022-0012; Yichang, Hubei, China). They were raised in separate cages in a specific pathogen free environment at 24 ± 2 °C with a 12 h/12 h light/dark cycle, acclimatized for one week, then divided randomly into 8 groups with 10 mice in each group (Table 1) (Lee et al., 2020). Group M1 received a normal diet, and Group M2–M4 had HFD (Table S1). Group M3 and Group M4 were given 25 and 50 mg/kg TSPJ (in 0.5 wt% sodium carboxymethyl cellulose, CMC-Na) by gavage daily, and Group M1 and Group M2 received the equivalent amount of 0.5 wt% CMC-Na.

For Group M5–M8, all mice had HFD and a daily gavage of 2.5 wt% dextran sulfate sodium salt (DSS, MP Biomedicals, Irvine, CA, USA). On days 8 and 15 of feeding, they received an intravenous injection of the lentivirus containing the miR-463-5p inhibitor (5 × 10⁷ TU per mouse, Group M7 and Group M8) or an empty vector (LV-NC, Group M5 and Group M6) (Guo et al., 2023). Group M6 and Group M8 were given 50 mg/kg TSPJ (in 0.5 wt% CMC-Na) by gavage daily, and Group M5 and Group M7 received the equivalent amount of 0.5 wt% CMC-Na.

After 4 months of feeding, 8 mice in each group were euthanized by cervical dislocation after an intraperitoneal injection of pentobarbital

Table 1
Creation of NASH model with male C57BL/6J mice and derivation of primary mouse hepatic macrophages.

Group	Diet/ Process	TSPJ (mg/ (kg·d))	Lentivirus	Mouse	Primary mouse hepatic macrophage
ND	Normal diet	0	N/A	M1	P1
HFD	HFD	0	N/A	M2	P2
TSPJ-L	HFD	25	N/A	M3	P3
TSPJ-H	HFD	50	N/A	M4	P4 ^a
LV-NC	HFD + 2.5% DSS	0	LV-NC empty vector	M5	P5
LV-NC + TSPJ	HFD + 2.5% DSS	50	LV-NC empty vector	M6	P6
LV- miR463 inh.	HFD +2.5% DSS	0	LV-miR- 463-5p inhibitor	M7	P7
LV- miR463 inh.+ TSPJ	HFD +2.5% DSS	50	LV-miR- 463-5p inhibitor	M8	P8

^a Also used for cultivation in the presence of GW4869 (referred to as Group TSPJ-H + GW4869).

sodium. The liver and blood samples were collected and stored at −80 °C. The other two mice in each group were used to isolate primary mouse hepatic macrophages according to a previously described method (Hu et al., 2023; Li et al., 2016; Ying et al., 2017). Specifically, the mice were anesthetized with 0.2% urethane, and hepatic portal vein cannulation was performed under sterile conditions. The liver was perfused in situ with D-Hanks balanced salt solution until it turned beige or light brown. It was then digested with 0.075% collagenase type IV for 10 min. The liver was removed, minced, finely ground with a sterile syringe, and filtered through a 200-mesh sieve. Trypsin was then added, and the mixture was kept in a shaking water bath at 37 °C for 30 min before centrifugation at 300g and room temperature for 8 min. The debris was dissolved thoroughly in hemocyte lysis buffer (12 mL), then mixed with Gey's balanced salt solution (GBSS, 12 mL) in a 50 mL centrifuge tube. The tube was sealed with paraffin film, shaken well in a water bath at 37 °C for 4 min, and centrifuged at 50g and room temperature for 5 min. The precipitates were suspended in Percoll (4 mL), and the suspension was centrifuged at 500g and room temperature for 5 min. The supernatant was passed through a 220 nm membrane filter to give the primary hepatic macrophages, which were subsequently grown in DMEM with 20% exosome-depleted fetal bovine serum. The primary hepatic macrophages derived from Group M4 were divided into two subgroups (TSPJ-H and TSPJ-GW) and grown separately. When the cell growth density reached 70%, GW4869 (10 iM) was added to the TSPJ-GW group, and DMSO of the same volume was added to all other groups. The supernatant was collected after additional incubation for 48 h.

The above animal experiments, designed following the ARRIVE guidelines and the recommendations from the National Health Association of the United States, were approved by the Animal Protection and Utilization Management Committee of China Three Gorges University (Approval No. 2021010M).

2.3. Cell experiments

2.3.1. Isolation and identification of exosomes

The culture of the primary mouse hepatic macrophages was centrifuged at 10000g and 4 °C for 30 min to remove the debris, and after ultrafiltration of the supernatant, the liquid was centrifuged further at 120000g and 4 °C for 90 min. The pellets were resuspended in PBS and centrifuged once more at 120000g and 4 °C for 70 min. The remaining pellets were collected as exosomes and suspended in PBS (300 iL). The

morphology and size of the particles were determined by transmission electron microscopy (TEM) (Hitachi, Tokyo, Japan), and the specific proteins associated with the vesicles were detected by Western blot (Ma et al., 2023).

2.3.2. Exosome uptake assay

The AML12 mouse hepatocyte cell line was obtained from the Cell Resource Center of Shanghai Institutes for Biological Sciences, Chinese Academy of Sciences. The macrophage-derived exosomes were marked with the red, fluorescent dye PKH26 (Solarbio, Beijing, China) and co-cultured with AML12 cells for 12 h. After the medium was decanted, the cells were washed with PBS three times, fixed with 4% paraformaldehyde for 15 min, and washed with PBS three times. The immobilized cells were permeabilized with 0.2% Triton X-100 (Wuhan Kerui Biotechnology Co. Ltd., Wuhan, China) at room temperature for 10 min, washed with PBS three times in the dark, incubated with DAPI for 10 min, and washed with PBS three times. An anti-fade mounting media was then introduced, and the uptake of the exosomes by the AML12 cells was detected by reverse fluorescence microscopy (Nikon, Tokyo, Japan) (Wan et al., 2022).

2.3.3. Cell models of NASH

The RAW264.7 mouse monocyte/macrophage cell line was obtained from the Cell Resource Center of Shanghai Institutes for Biological Sciences, Chinese Academy of Sciences. The cells were grown in Dulbecco's Modified Eagle Medium (DMEM) containing FBS and penicillin/streptomycin at 37 °C with 5% CO₂. The stock solution of TSPJ (300 mg/mL) was prepared by dissolving the dry TSPJ powder in dimethyl sulfoxide. The TSPJ solution used in the experiments was prepared by diluting the stock solution with DMEM.

The cell models were created according to Table 2. The RAW264.7 cells in Group C5–C8 were infected with the LV-NC empty vector lentivirus or the miR-463-5p inhibitor lentivirus (5 × 10⁷ TU/mL, 10 iL), and puromycin (3 iL/mL) was added 48 h after the infection. All cells (except Group C1) were then exposed to TSPJ or DMSO for 24 h before they were treated with 1 iL/mL lipopolysaccharide (LPS, Sigma, St. Louis, MO, USA) and 200 iL/mL palmitic acid (PA, Sigma, St. Louis, MO, USA) for 24 h to establish a lipotoxic inflammatory model (Wang et al., 2020).

The media that was used to culture the RAW264.7 cells and the primary hepatic macrophages from the animal models were isolated and passed through a 0.22 iL membrane filter, and the resulting supernatant was then used to culture AML12 cells at 37 °C for 24 h. The AML12 cells

were also incubated in DMEM with the isolated exosomes at 37 °C for 24 h.

2.4. Characterizations

2.4.1. Determination of TG, T-CHO, ALT, and AST

The collected mouse blood was left to stand at room temperature until stratification occurred, and it was then centrifuged at 3000 rpm and room temperature for 10 min to separate the serum in upper layer. The serum TG, TCHO, ALT, and AST were determined using specific assay kits (Nanjing Jiancheng Bioengineering Institute, Nanjing, China) according to the manufacturer's instructions (Nong et al., 2023).

2.4.2. Tissue staining and immunohistochemical analysis

The liver tissue was fixed in 4% paraformaldehyde for 24 h and then embedded in paraffin. The paraffin blocks were sliced into 4 iL thick sections, deparaffinized using xylene, and dehydrated using an ethanol gradient. The sections were subsequently stained with hematoxylin and eosin (H&E, Sigma-Aldrich, Steinheim, Germany) (Wan et al., 2022).

In addition, the paraffin sections were deparaffinized, rehydrated, heated in a citrate buffer at 120 °C and 1.5 MPa for 12 min for antigen retrieval, and blocked with 5% BSA for 1 h. The sections were then incubated with the anti-PHD2 primary antibody at 4 °C overnight and with the corresponding secondary antibody at 37 °C in the dark for 1 h. Finally, they were stained using DAB chromogen (Servicebio, Wuhan, China) and counterstained with hematoxylin (Wan et al., 2022).

2.4.3. Oil red O staining

The treated AML12 cells were fixed with 4% paraformaldehyde for 30 min, washed with PBS three times, then stained with the oil red O dye (Sigma, St. Louis, MO, USA) at room temperature in the dark for 30 min. After the excess stain was washed off with 60% isopropyl alcohol, the absorbance of the stained cells at 510 nm was measured and photographed under a microscope (Zhu et al., 2019).

2.4.4. MTT assay

RAW264.7 cells were seeded in a 96-well plate, with 6 replicates (wells) for each set. Once the confluence reached approximately 60%, the original medium was discarded and replaced with diluted TSPJ solution (5–300 iL/mL in DMEM, 100 iL). For the control cells, the medium was replaced with fresh complete culture medium (100 iL). After cultivation for 24 h, the MTT solution (5 mg/mL, 10 iL, Biosharp, Beijing, China) was added to each well, and the cultivation was continued at 37 °C for 4 h. The liquid in each well was then decanted, and DMSO (150 iL) was added. The plate was shaken gently in the dark for 5 min, and the absorbance at 490 nm was measured using a microplate reader (Thermo Fisher Scientific, Waltham, MA, USA) (Ma et al., 2023).

2.4.5. Immunofluorescence analysis

The primary mouse hepatic macrophages were treated sequentially with 4% paraformaldehyde for 30 min, 1% Triton for 15 min, and 5% BSA for 1 h, all of which were carried out at 25 °C. The cells were then incubated with the primary antibody (1:100, 200 iL) at 4 °C overnight and with the fluorescent secondary antibody (1:400) at 37 °C in the dark for 2 h. Finally, the nucleus was stained with DAPI, and photographs were captured using a reverse fluorescence microscope (Nikon, Tokyo, Japan) (Ma et al., 2023).

2.4.6. Western blot assay

Tissues or cells were digested using RIPA buffer, placed on ice for 10 min, then vortexed for 1 min. This procedure was repeated two more times. The total lysates were centrifuged at 12000g and 4 °C for 15 min. The protein contents were quantified by the bicinchoninic acid (BCA) assay. The protein lysates were subjected to 8%–10% sodium dodecyl-sulfate polyacrylamide gel electrophoresis (SDS-PAGE), and the bands were transferred onto polyvinylidene fluoride (PVDF) membranes. The

Table 2
Creation of *in vitro* NASH model with RAW264.7 macrophages.

Group	TSPJ (iL/mL) ^a	LPS + PA ^{b, a}	Lentivirus ^a	RAW264.7 macrophages
NC	0 ^c	None ^b	N/A	C1
LPS/PA	0 ^c	+	N/A	C2
LPS/PA + TSPJ-L	15	+	N/A	C3
LPS/PA + TSPJ-H	30	+	N/A	C4
LV-NC	0 ^c	+	LV-NC empty vector	C5
LV-NC + TSPJ	30	+	LV-NC empty vector	C6
LV-miR463 inh.	0 ^c	+	LV-miR-463-5p inhibitor	C7
LV-miR463 inh. + TSPJ	30	+	LV-miR-463-5p inhibitor	C8

^a The conditions describe the treatment of RAW264.7 macrophages. The culture media were collected for the extraction of exosomes and for the incubation of AML12 cells.

^b LPS, 1 iL/mL lipopolysaccharide; PA, 200 iL/mL palmitic acid. Group C1 was treated with DMEM for 24 h.

^c Exposed to the equivalent amount of DMSO.

membranes were blocked with 5% skim milk at 4 °C for 2 h and then incubated overnight at 4 °C with the following primary antibodies: FASN (1:2000, ABclonal, Wuhan, China), ACC1 (1:2000, ABclonal, Wuhan, China), SREBP-1C (1:2000, Wanleibio, Shenyang, China), TNF- α (1:1000, Wanleibio, Shenyang, China), IL-1 α (1:2000, Wanleibio, Shenyang, China), Ym-1 (1:2000, Abcam, Cambridge, UK), IL-10 (1:1000, Wanleibio, Shenyang, China), CD9 (1:500, Santa Cruz, Dallas, TX, USA), Syntenin-1 (1:750, Santa Cruz, Dallas, TX, USA), TSG101 (1:1000, Santa Cruz, Dallas, TX, USA), HSP70 (1:1000, Santa Cruz, Dallas, TX, USA), and PHD2 (1:2000, ABclonal, Wuhan, China). The membranes were then washed with TBST buffer three times and treated with the secondary antibodies at room temperature for 1 h. The protein bands were visualized using an electrochemiluminescence imaging system (Bio-rad, Hercules, CA, USA) and quantitatively analyzed using ImageJ. β -Actin (1:100000, ABclonal, Wuhan, China) was used as the internal reference (Ma et al., 2023).

2.4.7. Reverse transcription quantitative polymerase chain reaction (RT-qPCR)

The total RNA was extracted from tissues and cells using TRIzol (Vazyme, Nanjing, China), and the cDNA was synthesized using the HiScript III 1st Strand cDNA Synthesis Kit (Vazyme, Nanjing, China). RT-qPCR was then conducted using a 7900HT qPCR system (Applied Biosystems, Waltham, MA, USA) and the SYBR Green Master Mix (Vazyme, Nanjing, China) with initial denaturation at 95 °C for 10 min followed by 40 amplification cycles of 95 °C for 5 s and 60 °C for 30 s. The results were quantified using the $2^{-\Delta\Delta Ct}$ method, and β -actin and U6 were used as the internal control for the detection of mRNA and miRNA, respectively. The primer sequences used in the experiments are listed in Table S2 (Hu et al., 2023).

2.4.8. Dual-luciferase reporter assay

The luciferase reporter assay was conducted using the PmirGLO luciferase reporter vector (TransGen Biotech, Wuhan, China). After AML12 cells were co-transfected with either wild-type or mutant PHD2 3' UTR plasmids along with miR-463-5p inhibitor or control, the cells were lysed with $1 \times$ universal lysis buffer. Subsequently, the supernatant from the cell lysate (20 μ L) was mixed with the luciferase substrate (100 μ L), and the firefly luciferase activity was measured using a luminometer immediately. The Renilla substrate (100 μ L) was then added to evaluate Renilla luciferase activity (Niculescu et al., 2018).

2.5. Bioinformatic analysis

The downstream targets of mmu-miR-463-5p were identified utilizing miRDB, TargetScan, and miWalk databases. The results from these databases were visualized using a Venn diagram.

2.6. Statistical analysis

Statistical analyses were conducted using SPSS 20.0. Data were shown as the mean $\pm$ standard error of the mean (SEM) and plotted using GraphPad Prism 7.0. The differences between groups were assessed using Student's *t*-test when the data were normally distributed and using the Mann-Whitney *U* test when otherwise. Differences were deemed statistically significant when $P < 0.05$.

3. Results

3.1. TSPJ alleviated lipid deposition and inflammatory response in the liver of HFD-induced NASH mice

A NASH model was established utilizing HFD-induced C57BL/6J mice to investigate the effects of TSPJ on hepatic lipid metabolism. After 4 months of feeding, TSPJ effectively reduced the body weight, liver weight, and the ratio of liver weight to body weight in HFD-induced

mice (Fig. 1A). According to the H&E staining results, the hepatic steatosis and vacuolization in HFD-fed mice were significantly alleviated by TSPJ intervention (Fig. 1B). The TSPJ treatment also decreased the serum levels of TG, T-CHO, ALT, and AST (Fig. 1C). Regarding the lipid metabolism and the inflammatory responses of the liver, the TSPJ treatment reduced the expression of lipid synthesis-related genes, including *FASN*, *ACC1*, and *SREBP-1C* (Fig. 1D), as well as the gene expression of inflammatory cytokines TNF α and IL-1 α (Fig. 1E). These changes tended to be more pronounced upon administration of TSPJ at a higher dose (TSPJ-H). It can be concluded that TSPJ effectively alleviated hepatic lipid deposition and inflammatory responses in HFD-fed mice.

3.2. TSPJ inhibited the M1 polarization and promoted the M2 polarization of macrophages

The impact of TSPJ on the viability of RAW264.7 macrophage was evaluated using the MTT assay. It was found that when the concentration fell within 5–175 μ g/mL, TSPJ had no noticeable effect on the viability of RAW264.7 cells and there was no obvious cytotoxicity (Fig. 2A). Subsequently, 15 and 30 μ g/mL were selected as the doses of TSPJ for further *in vitro* studies, since the cell viability was the highest at these two concentration levels. To examine the impact of TSPJ on macrophage polarization, a lipotoxic inflammation model was created by inducing RAW264.7 cells with LPS and PA. Both RT-qPCR and Western blot analyses demonstrated that LPS and PA significantly upregulated M1 markers (TNF α and IL-1 α) and downregulated M2 markers (Ym-1 and IL-10) in RAW264.7 cells. However, after TSPJ intervention, there was a significant downregulation of M1 markers (Fig. 2B) and a notable upregulation of M2 markers (Fig. 2C). Therefore, TSPJ inhibited the M1 polarization and promoted the M2 polarization of macrophages.

3.3. TSPJ reduced the metabolic disorder in hepatocytes through macrophages

Hepatic macrophages can disrupt lipid metabolism in hepatocytes by regulating the expression of genes associated with lipid metabolism and releasing inflammatory factors (Kazankov et al., 2019). The culture media of the lipotoxic inflammation model, created by inducing RAW264.7 cells with LPS/PA, were collected by ultrafiltration and used to cultivate AML12 hepatocytes. The oil red O staining showed that when the macrophages were induced with LPS/PA, the AML12 cells had numerous aggregated red lipid droplets compared to the control, and the absorbance at 510 nm after the isopropanol wash was significantly higher (Fig. 3A). If the RAW264.7 cells were treated with TSPJ, there were fewer lipid droplets in the AML12 hepatocytes, and the reduction was the most significant when the dose of TSPJ was higher. The RT-qPCR analysis showed that the mRNA expression levels of *FASN*, *ACC1*, and *SREBP-1C* were significantly elevated in the AML12 cells when the macrophages were induced by LPS/PA but markedly reduced when the macrophages were protected by TSPJ (Fig. 3B). The Western blot analysis had similar results (Fig. 3C). Thus, the lipid deposition in hepatocytes was alleviated when the LPS/PA-treated macrophages were protected by TSPJ.

Although RAW264.7 cells are a widely used macrophage cell line with stable and reproducible characteristics, they may not fully represent the complex behavior of primary macrophages in their native tissue environment. To capture the full spectrum of macrophage behavior in a physiological setting, primary mouse hepatic macrophages were isolated and cultured to further validate how hepatic macrophages mediated the effect of TSPJ on hepatic lipid deposition. The immunofluorescence staining showed that the isolated primary cells were positive for F4/80 and weakly positive for CD11b, which confirmed the successful extraction (Fig. 3D). Subsequently, AML12 cells were treated with the culture media of the primary mouse hepatic

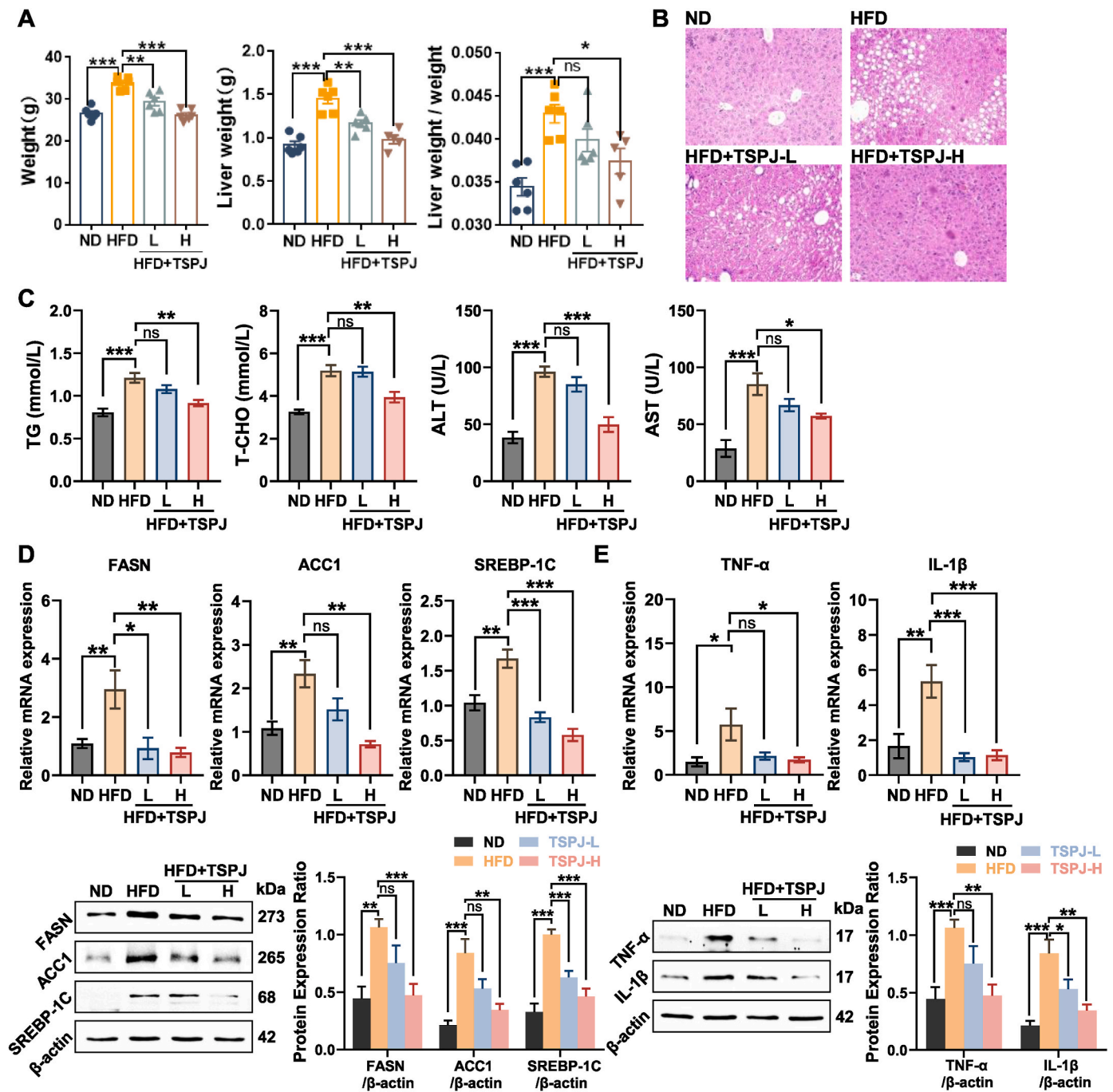


Fig. 1. Effects of TSPJ on lipid metabolism and inflammation in the liver tissue of HFD-induced NASH mice. (A) Changes in body weight, liver weight, and ratio of liver weight to body weight. (B) Representative H&E staining images of the liver. (C) The serum levels of TG, T-CHO, ALT, and AST. Hepatic mRNA and protein expression of (D) FASN, ACC1, SREBP-1C, and (E) TNF-α and IL-1β. Data are expressed as mean ± SEM (n = 6–8). **P* < 0.05; ***P* < 0.01; ****P* < 0.001; ns, not significant.

macrophages. The oil red O staining (Fig. 3E), RT-qPCR (Fig. 3F), and Western blot (Fig. 3G) results of the AML12 cells were consistent with the prior findings. The AML12 cells exhibited stronger lipid accumulation and an upregulation in the lipid synthesis-related genes if the mice were given HFD, but these trends were weakened if the mice received TSPJ at the same time. However, if the exosome secretion inhibitor GW4869 was added during the cultivation of the primary hepatic macrophages, the AML12 cells exhibited significantly increased the formation of lipid droplets as well as enhanced expression of FASN, ACC1, and SREBP-1C at both mRNA and protein levels. Therefore, TSPJ reduced the hepatic lipid accumulation via hepatic macrophages, and

exosomes must have mediated how hepatic macrophages influenced the hepatic lipid deposition.

3.4. TSPJ decreased lipid deposition in hepatocytes via the exosomes from hepatic macrophages

Exosomes were extracted from the primary mouse hepatic macrophages. On the TEM images, the exosomes were cup-shaped particles about 30–150 nm in size (Fig. 4A). According to nanoparticle tracking analysis, the average diameter of the exosomes was 132.1 nm, and the particle concentration was 4.1×10^{11} particles/mL (Fig. 4B). The

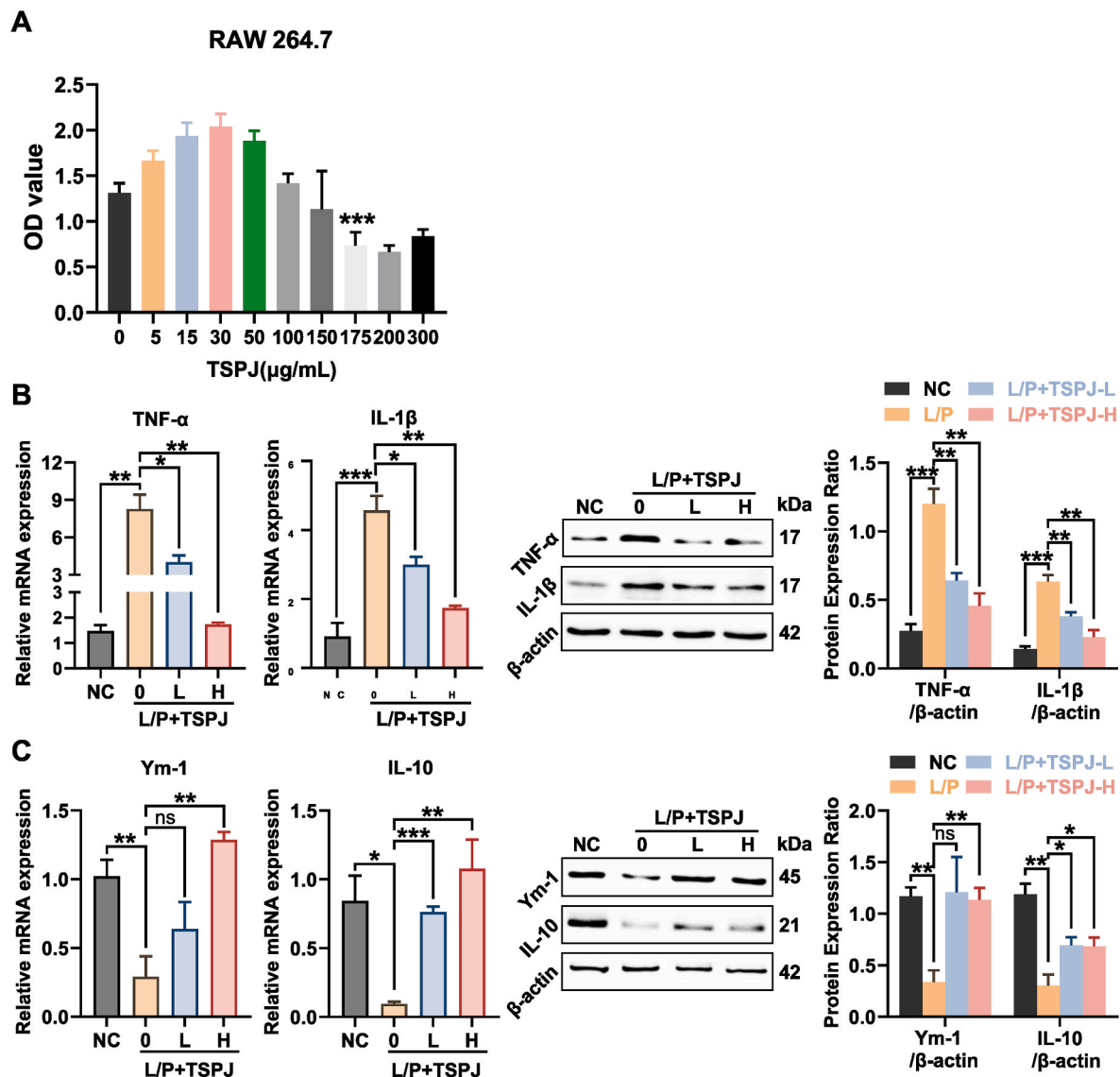


Fig. 2. Effects of TSPJ on macrophages. (A) MTT assay of the cell viability of RAW264.7 cells after treatment with TSPJ at different concentrations for 24 h. (B, C) The mRNA and protein expressions of TNF- α , IL-1 β , Ym-1, and IL-10 in RAW264.7 cells after treatment with LPS/PA and TSPJ for 24 h. Data are expressed as mean $\pm$ SEM (n = 4–6). * P < 0.05; ** P < 0.01; *** P < 0.001; ns, not significant.

expression of the exosome marker proteins CD9, Syntenin-1, TSG101, and HSP70 in the isolated exosomes was confirmed by Western blot analysis (Fig. 4C). Immunofluorescence staining showed that PKH26-labeled exosomes could enter the AML12 hepatocytes (Fig. 4D). Subsequently, the AML12 hepatocytes were incubated with the exosomes derived from different sources, and the lipid metabolism in the hepatocytes was analyzed by oil red O staining (Fig. 4E), RT-qPCR (Fig. 4F), and Western blot (Fig. 4G). When the exosomes were derived from the mice receiving HFD, the AML12 cells exhibited stronger lipid droplet formation and an upregulation of FASN, ACC1, and SREBP-1C. However, these changes were attenuated if the mice received TSPJ along with HFD. Therefore, TSPJ must have modulated lipid metabolism in hepatocytes via the exosomes derived from hepatic macrophages.

3.5. TSPJ increased the expression of miR-463-5p in macrophages and exosomes

The differential expression of miRNAs in the liver of the mice was evaluated using RT-qPCR. The expression of miR-463-5p in the liver tissue of HFD mice was significantly reduced, but it was markedly

increased after the mice were treated with TSPJ. The expression levels of miR-935, miR-206-3p, and miR-136-5p did not show any statistically significant difference between the mice receiving a normal diet and those receiving HFD with or without TSPJ (Fig. 5A). Similar changes in the miR-463-5p expression were observed in the exosomes derived from the RAW 264.7 macrophages and the primary hepatic macrophages (Fig. 5B). The miRDB, miRWalk, and TargetScan databases were employed to predict the potential targets of miR-463-5p, and *PHD2* emerged as one of the six candidate genes (Fig. 5C). Eight complementary base pairing sequences existed between miR-463-5p and *PHD2* according to the TargetScan database (Fig. 5D), and previous studies have shown that the miR-463-5p mimic downregulates *PHD2*. Thus, miR-463-5p likely regulated the expression of *PHD2* via the 3' UTR. To verify the direct and specific interaction between the miR-463-5p and *PHD2*, both wild-type and mutant *PHD2* were tested in the luciferase assay. The results showed that miR-463-5p significantly inhibited the luciferase activity of wild-type *PHD2* but had no effect on the mutant *PHD2* (Fig. 5E). Thus, miR-463-5p specifically bound to the 3' UTR of *PHD2* to post-transcriptionally downregulate the expression of *PHD2*.

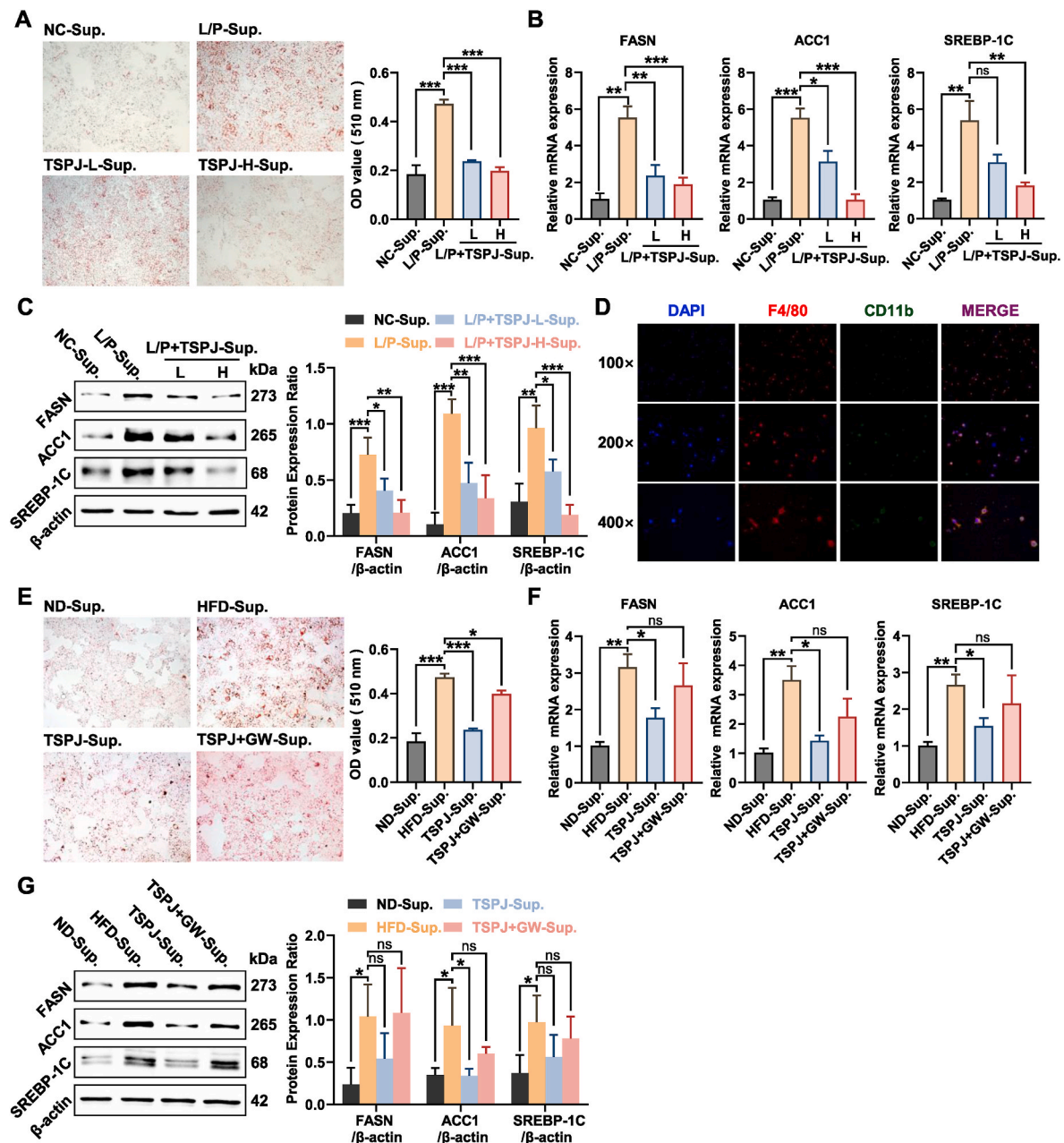


Fig. 3. TSPJ intervention alleviated the lipid metabolism disorder of liver cells through macrophages. (A) AML12 cells treated with the culture medium of RAW264.7 cells: Representative oil red O staining images and OD values. (B, C) The mRNA and protein expression of FASN, ACC1, and SREBP-1C in AML12 cells. (D) Identification of primary hepatic macrophages by immunofluorescence staining. (E) AML12 cells treated with the culture medium of primary hepatic macrophages: Representative oil red O staining images and OD values. (F, G) The mRNA expression of lipid synthesis-related genes and the levels of relevant proteins in AML12 cells. Data are expressed as mean $\pm$ SEM ($n = 4$). * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; ns, not significant. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

3.6. TSPJ inhibited PHD2 expression in hepatocytes through macrophage-derived exosomes

PHD2 is a gene that regulates cellular responses to hypoxia, and a lower PHD2 expression can alleviate HFD-induced obesity and related lipid metabolism abnormalities (Laitakari et al., 2019). In the mouse models, HFD significantly upregulated PHD2 in the liver at both mRNA and protein levels, but TSPJ reduced PHD2 in a dose-dependent manner (Fig. 6A and B). The immunohistochemical staining results also confirmed that TSPJ inhibited the expression of PHD2 in the liver (Fig. 6C).

When the AML12 cells were incubated with the medium that was

used to culture RAW267.4 macrophages, the mRNA expression and the protein level of PHD2 were both elevated if the macrophages were induced with LPS/PA, but the rise in the PHD2 levels of the AML12 cells was abated if the macrophages were also exposed to TSPJ (Fig. 6D and E). The same results were observed from the immunofluorescence experiments (Fig. 6F).

When the AML12 cells were incubated with the culture medium of the primary mouse hepatic macrophages, the mRNA expression and protein level of PHD2 increased significantly when the mice received HFD, decreased significantly if the mice also received a high dose of TSPJ, and increased significantly if the exosome inhibitor GW4869 was supplemented (Fig. 6G and H). When the AML12 cells were cultured

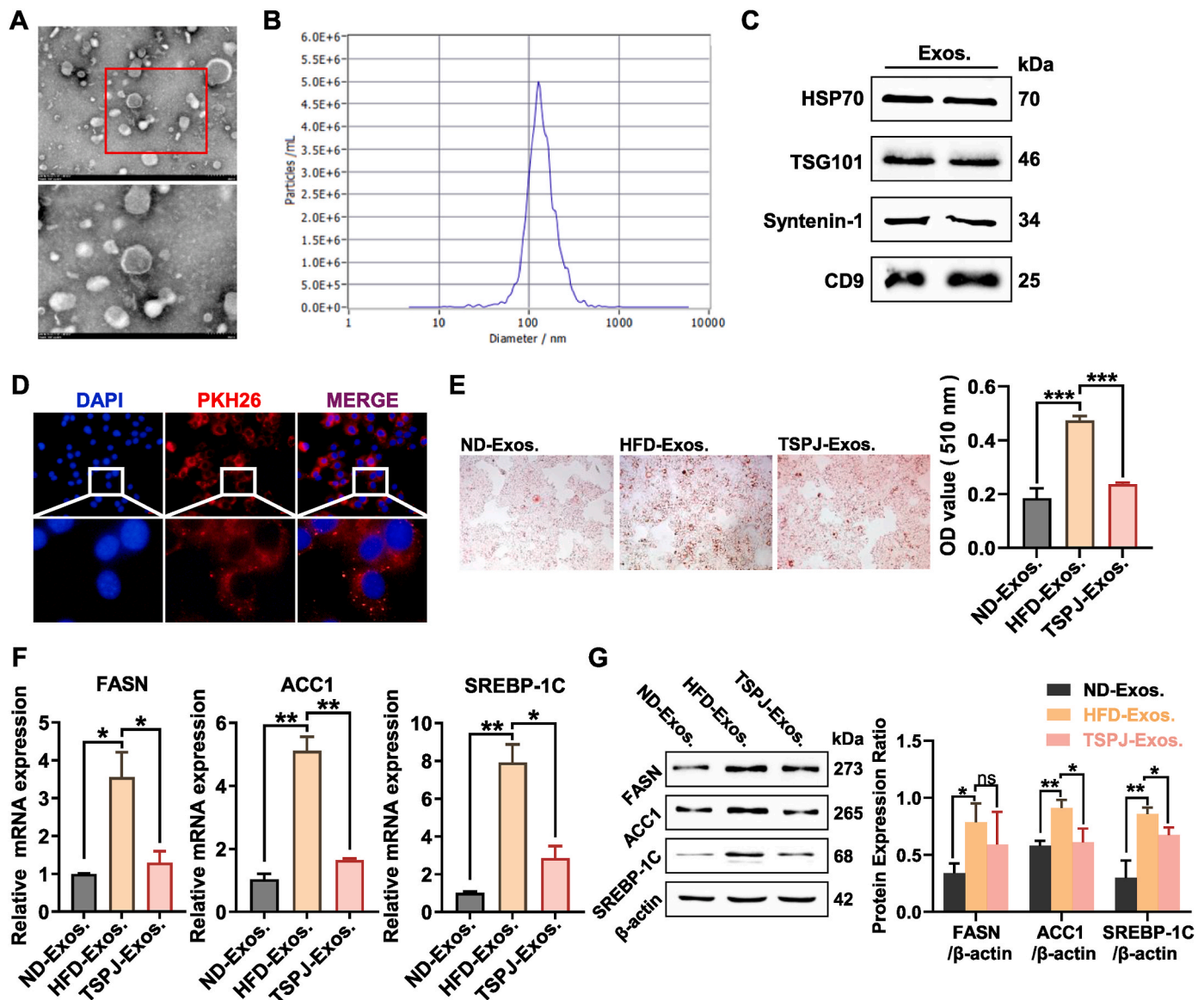


Fig. 4. Exosomes mediated how TSPJ regulated lipid deposition. (A) Representative transmission electron microscopic photographs of liver macrophage-derived exosomes. (B) Particle size analysis of exosomes. (C) Expression of exosome marker proteins. (D) The uptake of exosomes by liver cells detected by immunofluorescence staining. (E) AML12 cells incubated with exosomes from different sources: Representative oil red O staining images and OD values. (F, G) The mRNA and protein expressions of FASN, ACC1, and SREBP-1C in AML12 cells. Data are expressed as mean $\pm$ SEM ($n = 4$). * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$, ns: not significant. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

with the isolated exosomes, the mRNA expression and protein level of PHD2 were elevated if the mice received HFD but decreased if the mice also received TSPJ (Fig. 6I and J). Thus, TSPJ must have inhibited PHD2 expression in hepatocytes through the exosomes derived from hepatic macrophages.

3.7. The benefits of TSPJ were offset by miR-463-5p knockdown

To further evaluate the crucial role of miR-463-5p, mouse models and RAW264.7 cell models with miR-463-5p knockdown were created. For both the mice and the cells, the transfection of the lentivirus carrying the miR-463-5p inhibitor significantly reduced the expression of miR-463-5p (Fig. 7A). For the mice transfected with the negative control of the lentivirus, TSPJ significantly reduced the body weight and liver weight. For those mice that took TSPJ, the miR-463-5p knockdown increased the body weight and liver weight (Fig. 7B). That is, TSPJ and miR-463-5p knockdown must have influenced the body weight and liver

weight of the HFD-induced mice in the opposite direction. Fig. 7C shows the images of the liver tissue of the mouse models. For the mice transfected with the negative control of the lentivirus, the liver had a yellowish color and had blunt edges if the mice did not receive TSPJ, but it returned to a normal size with reddish color and soft texture when the mice received TSPJ. For the mice with miR-463-5p knockdown, the liver had lighter color, larger size, and blunt edges regardless of their TSPJ intake. For the mice that did not receive TSPJ, there were disordered liver cords and increased lipid vacuoles with inflammatory cell infiltration in the H&E staining image of the liver. Once the mice received TSPJ, the cells became more tightly and regularly arranged, and the volume and number of lipid vacuoles both decreased. Nevertheless, if the mice had miR-463-5p knockdown, the cells were more disordered and had larger lipid vacuoles even if the mice also took TSPJ (Fig. 7D).

In the absence of miR-463-5p knockdown, TSPJ reduced the serum levels of TG, T-CHO, ALT, and AST in HFD mice. The miR-463-5p knockdown increased the serum levels of TG, T-CHO, ALT, and AST in

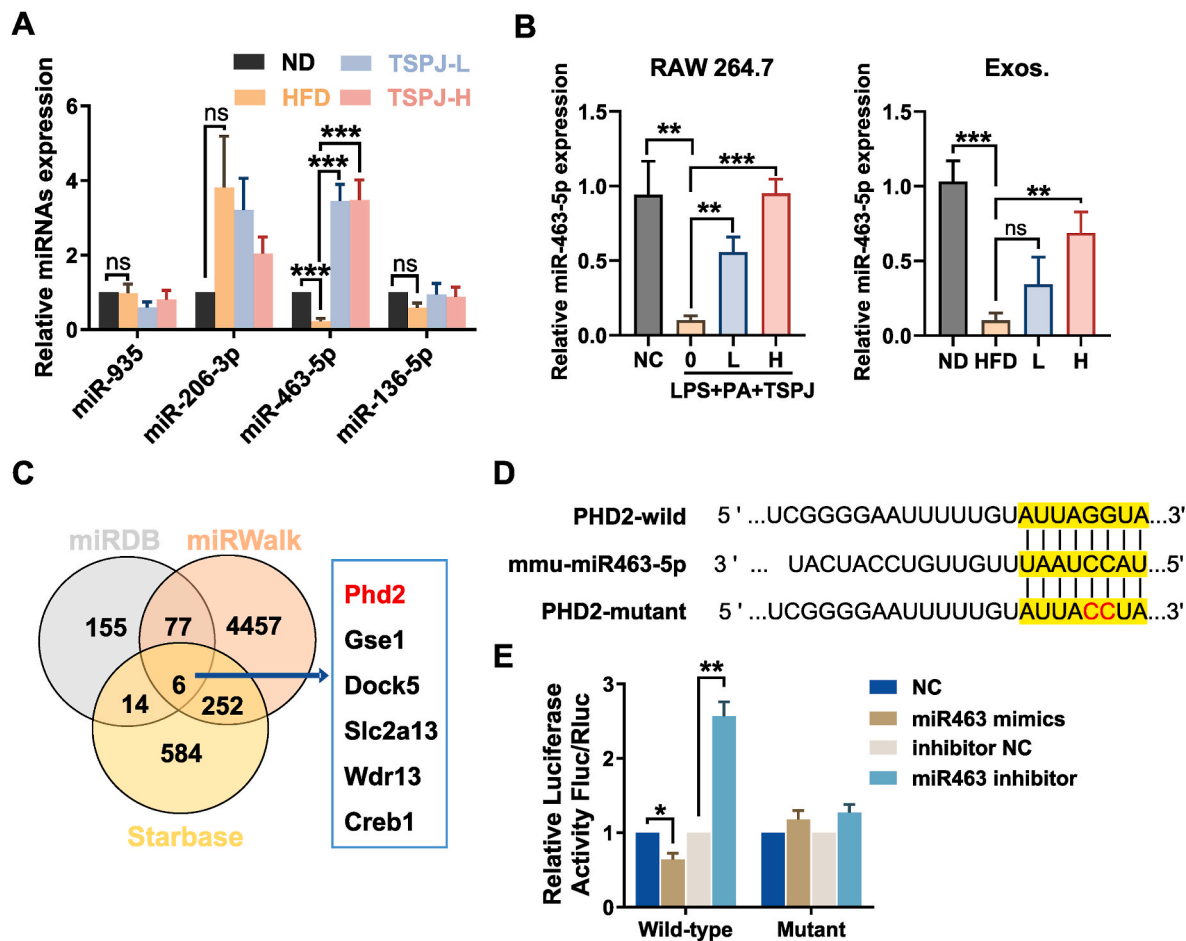


Fig. 5. Regulatory relationships between TSPJ, miR-463-5p, and PHD2. (A) Quantification of the expression of different miRNAs in mouse liver using RT-qPCR. (B) The expression of miR-463-5p in RAW264.7 cells and exosomes from the primary hepatic macrophages. (C) Prediction of the downstream target genes of miR-463-5p. (D) The binding sites between miR-463-5p and PHD2. (E) Dual-luciferase reporter assay of wild-type and mutant PHD2. Data are expressed as mean $\pm$ SEM (n = 4–8). * P < 0.05; ** P < 0.01; *** P < 0.001; ns, not significant.

HFD mice regardless of TSPJ (Fig. 7E). In the presence of the miR-463-5p knockdown, the mRNA expression and protein levels of FASN, ACC1, and SREBP-1C were no longer reduced markedly even when the mice took TSPJ (Fig. 7F and G). The miR-463-5p knockdown also sabotaged TSPJ in controlling the expression of the inflammatory factors TNF α and IL-1 α (Fig. 7H and I), and it increased the mRNA expression and protein levels of PHD2 in the liver (Fig. 7J and K). Taken together, it could be inferred that TSPJ alleviated HFD-induced lipid metabolism disorders in mice by upregulating miR-463-5p to inhibit PHD2 expression.

3.8. TSPJ downregulated PHD2 in hepatocytes via exosomal miR-463-5p

In the RAW264.7 cells, the miR-463-5p knockdown increased the levels of TNF α and IL-1 α and decreased the levels of Ym-1 and IL-10 (Fig. 8A and B). Thus, miR-463-5p must be able to induce the M2 polarization and inhibit the M1 polarization of macrophages. The AML12 cells were then incubated using the supernatant of the macrophage culture. According to oil red O staining, the AML12 cells had less accumulation of lipid droplets when the macrophage was treated with TSPJ (Fig. 8C). The mRNA expression and protein levels of FASN, ACC1, and SREBP-1C in the AML12 cells were also reduced if the macrophage was treated with TSPJ (Fig. 8D and E). The miR-463-5p knockdown in the macrophage, however, counteracted TSPJ in regulating the lipid accumulation and the expression of lipid synthesis-related genes in the AML12 cells. In addition, the PHD2 levels in the AML12 cells were suppressed if the macrophage was treated with TSPJ and increased if the

macrophage had miR-463-5p knockdown (Fig. 8F–H).

Subsequently, AML12 cells were cultured with the exosomes isolated from the primary mouse hepatic macrophages. When the mice were transfected with the negative control of the lentivirus, the AML12 cells had abundant mature lipid droplets, and the number of lipid droplets in the AML12 cells decreased when the mice received TSPJ. The AML12 cells had more, larger, and clustered lipid droplets when the mice had miR-463-5p knockdown regardless of TSPJ intake (Fig. 8I). The miR-463-5p knockdown in the mice was also associated with the upregulation of lipid synthesis-related genes and the increased expression of PHD2 at both mRNA and protein levels in the AML12 cells (Fig. 8J–L). Thus, TSPJ must have upregulated miR-463-5p in the hepatic macrophage-derived exosomes, which then suppressed PHD2 expression in hepatocytes to alleviate the lipid metabolism disorders and inflammatory responses in the hepatocytes.

4. Discussion

NASH is an aggressive form of NAFLD characterized by inflammation, hepatocellular steatosis, and varying degrees of fibrosis (Sheka et al., 2020). The progression of NAFLD is not determined solely by individual cells but is heavily influenced by the interactions and cross-talk between cells (Garcia-Tsao, 2021). The hepatic macrophages, which are primarily derived from KCs and monocytes, are particularly important in mediating inflammatory responses and lipid metabolism disorders (Guilliams and Scott, 2022). During the progression of NAFLD,

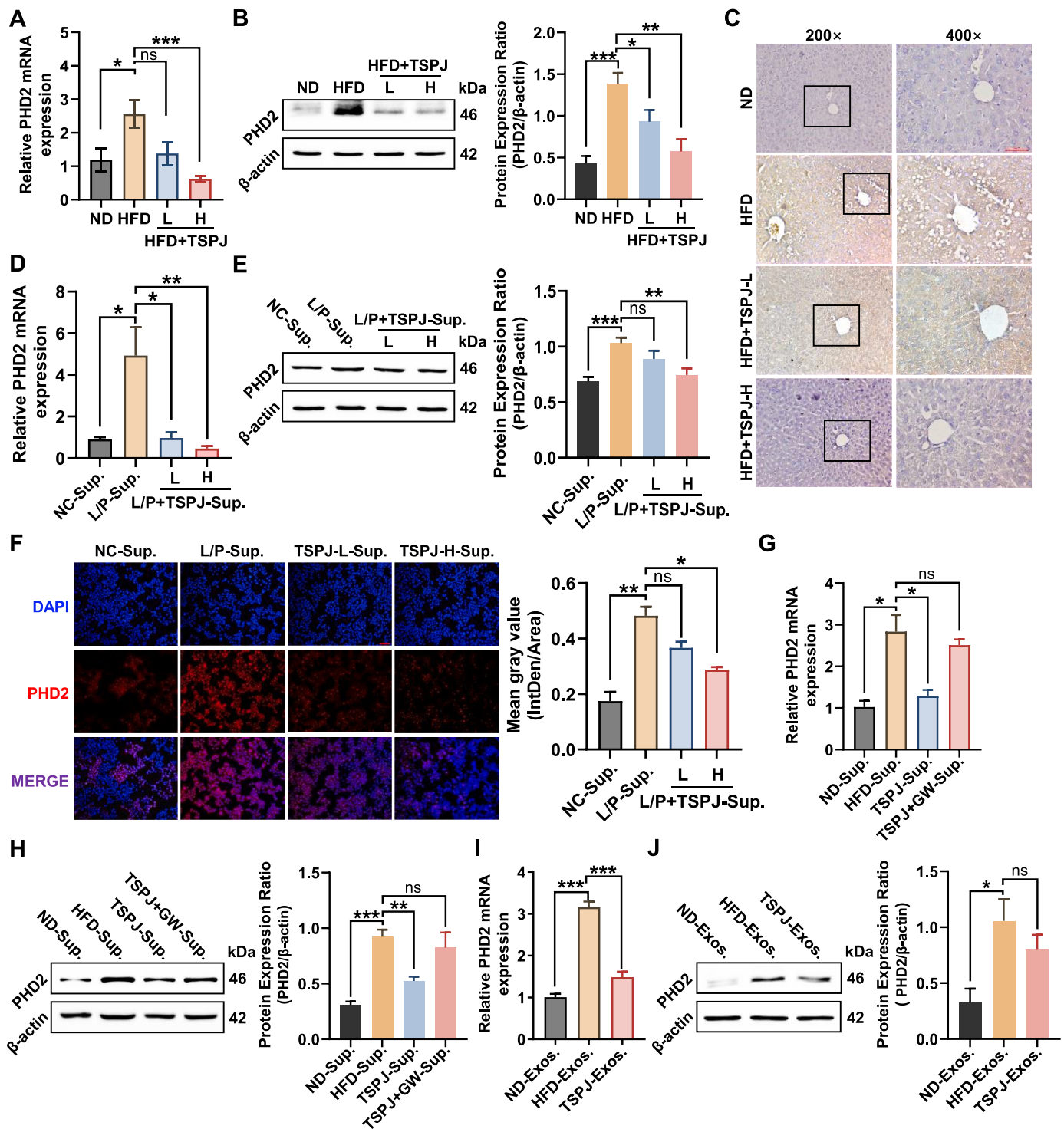


Fig. 6. TSPJ inhibited PHD2 expression in hepatocytes through macrophage-derived exosomes. (A, B) The effects of TSPJ on the mRNA and protein expression of PHD2 in the liver tissue. (C) Representative immunohistochemical staining images. (D) Western blot, (E) RT-qPCR, and (F) immunofluorescence staining analysis of the effects of TSPJ on PHD2 expression in AML12 cells. (G, H) The effects of TSPJ and GW4869 on the mRNA and protein expressions of PHD2. (I, J) The effects of exosomes derived from hepatic macrophages on the mRNA and protein expressions of PHD2 in AML12 cells. Data are expressed as mean $\pm$ SEM ($n = 4-8$). * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; ns, not significant.

hepatocytes can regulate the differentiation and recruitment of macrophages, and the activated hepatic macrophages can regulate hepatocytes via exosomes to accelerate or slow the progression of NAFLD (Kazankov et al., 2019). For example, exosome-derived miR-396e regulates PFKFB3 expression through the macrophage-hepatocyte crosstalk to attenuate

NAFLD (Liu et al., 2023). Unfortunately, effective therapeutic options for NAFLD are currently extremely limited. Traditional Chinese medicine, known for its multi-component nature and diverse mechanisms, demonstrates unique advantages in treating NAFLD (Zhou et al., 2022). In multiple studies, TSPJ, which are derived from the traditional Chinese

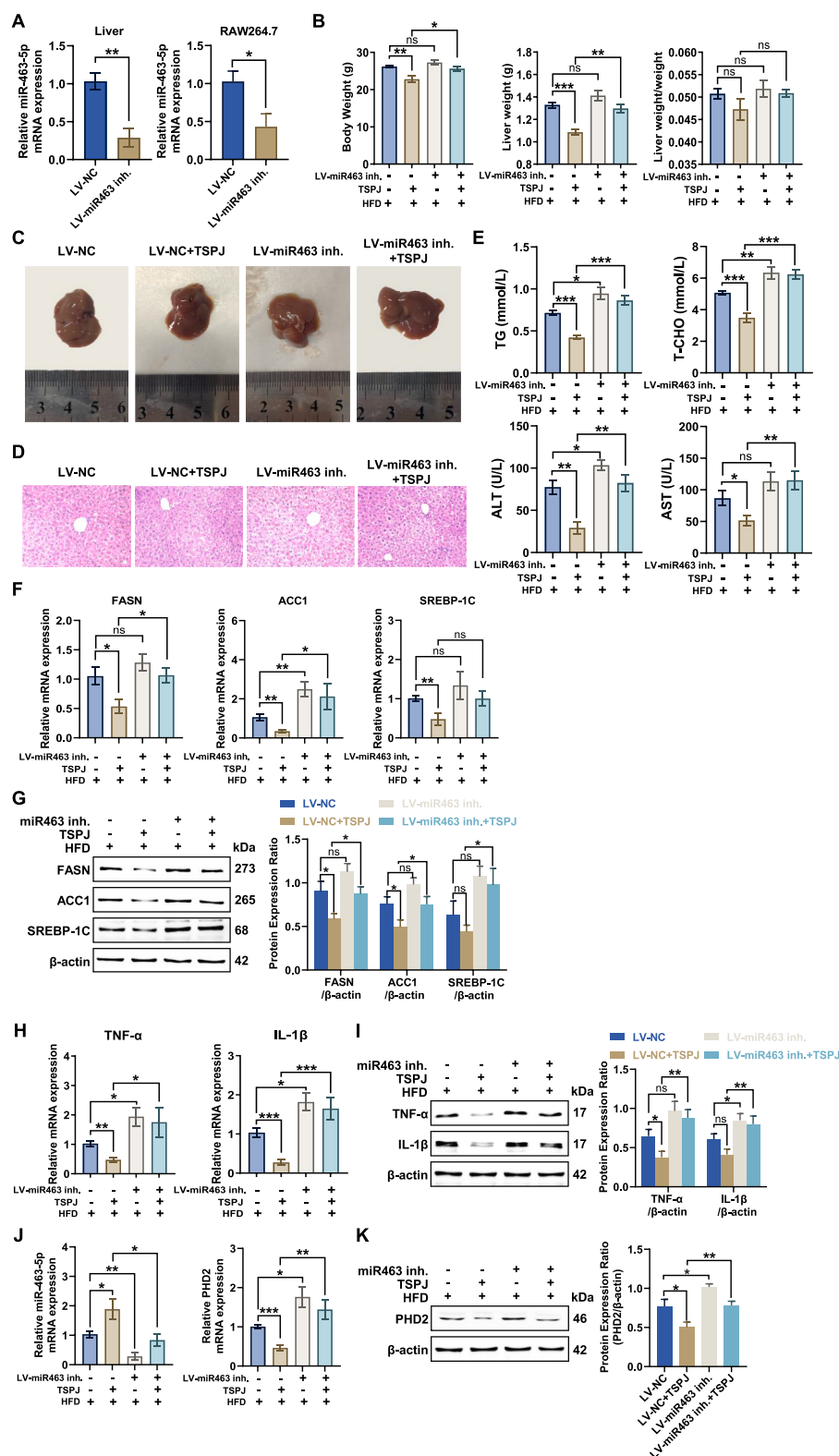


Fig. 7. miR-463-5p knockdown reduced the benefits of TSPJ. (A) The relative miR-463-5p expression in the liver tissue of mice and RAW264.7 cells. (B) The body weight, liver weight, and ratio of liver weight to body weight of different mouse models. (C) Liver morphology images and (D) H&E staining images of mouse models. (E) The levels of TG, T-CHO, ALT, and AST in serum of mice. The mRNA and protein expressions of (F, G) FASN, ACC1, and SREBP-1C, and (H, I) TNF- α and IL-1 β in the liver tissue of mouse models. (J) The expression of miR-463-5p, the mRNA expression of PHD2, and (K) the protein level of PHD2 in the liver tissue of mouse models. Data are expressed as mean $\pm$ SEM (n = 4–8). * P < 0.05; ** P < 0.01; *** P < 0.001; ns, not significant.

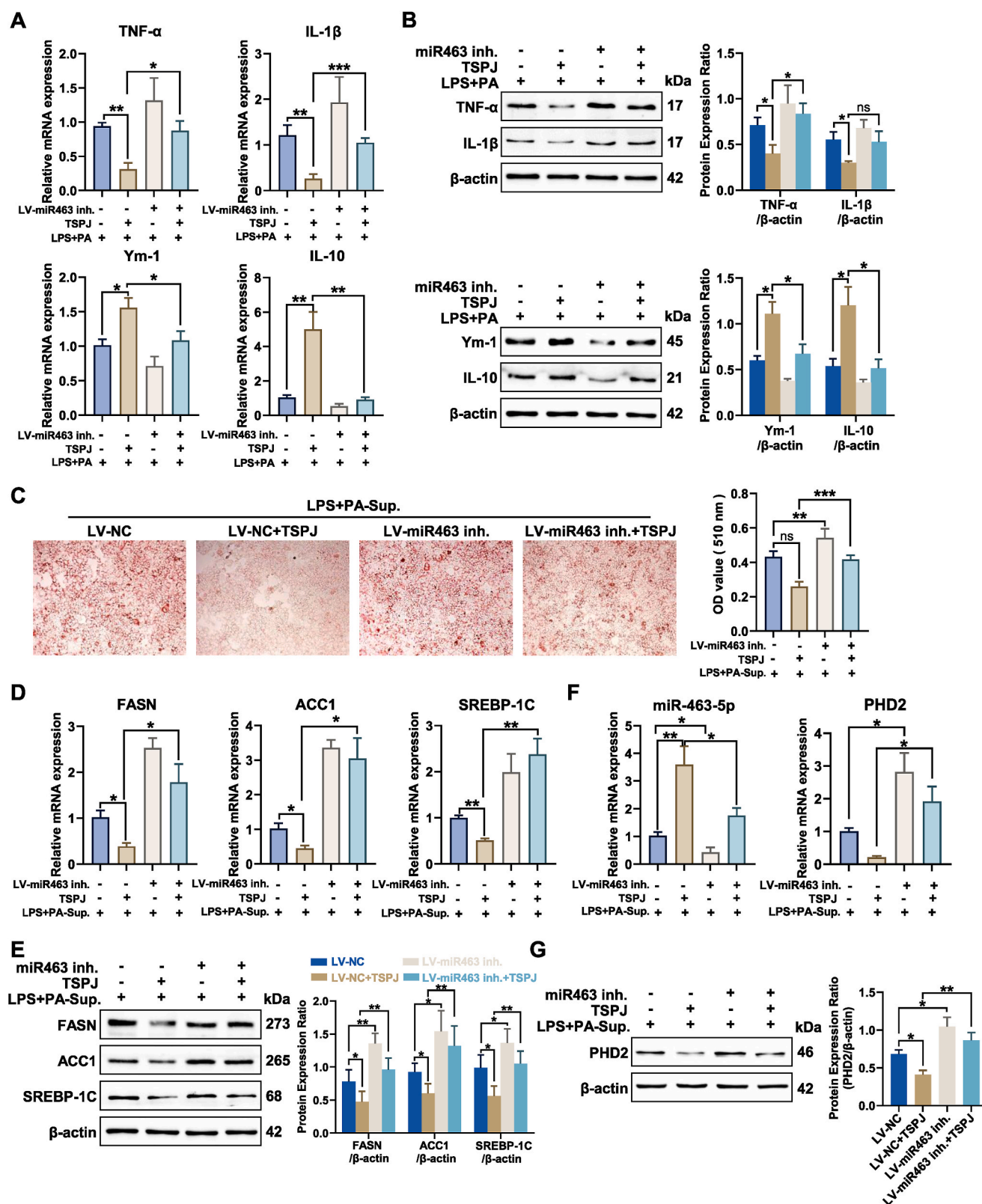


Fig. 8. TSPJ attenuated hepatocyte inflammation and lipid deposition by targeting PHD2 via exosomal miR-463-5p. (A, B) The mRNA and protein expressions of M1/M2 biomarkers. (C) AML12 cells treated with the culture medium of RAW264.7 cells having miR-463-5p knockdown: Representative oil red O staining images and OD values. (D–G) The expression of FASN, ACC1, SREBP-1C, miR-463-5p, and PHD2 in AML12 cells. (H) Representative immunofluorescence staining images of AML12 cells. (I) AML12 cells treated with the culture medium of primary hepatic macrophages having miR-463-5p knockdown: Representative oil red O staining images and OD values. (J–L) The expression of FASN, ACC1, SREBP-1C, miR-463-5p, and PHD2 in AML12 cells. Data are expressed as mean $\pm$ SEM ($n = 3-4$). * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; ns, not significant. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

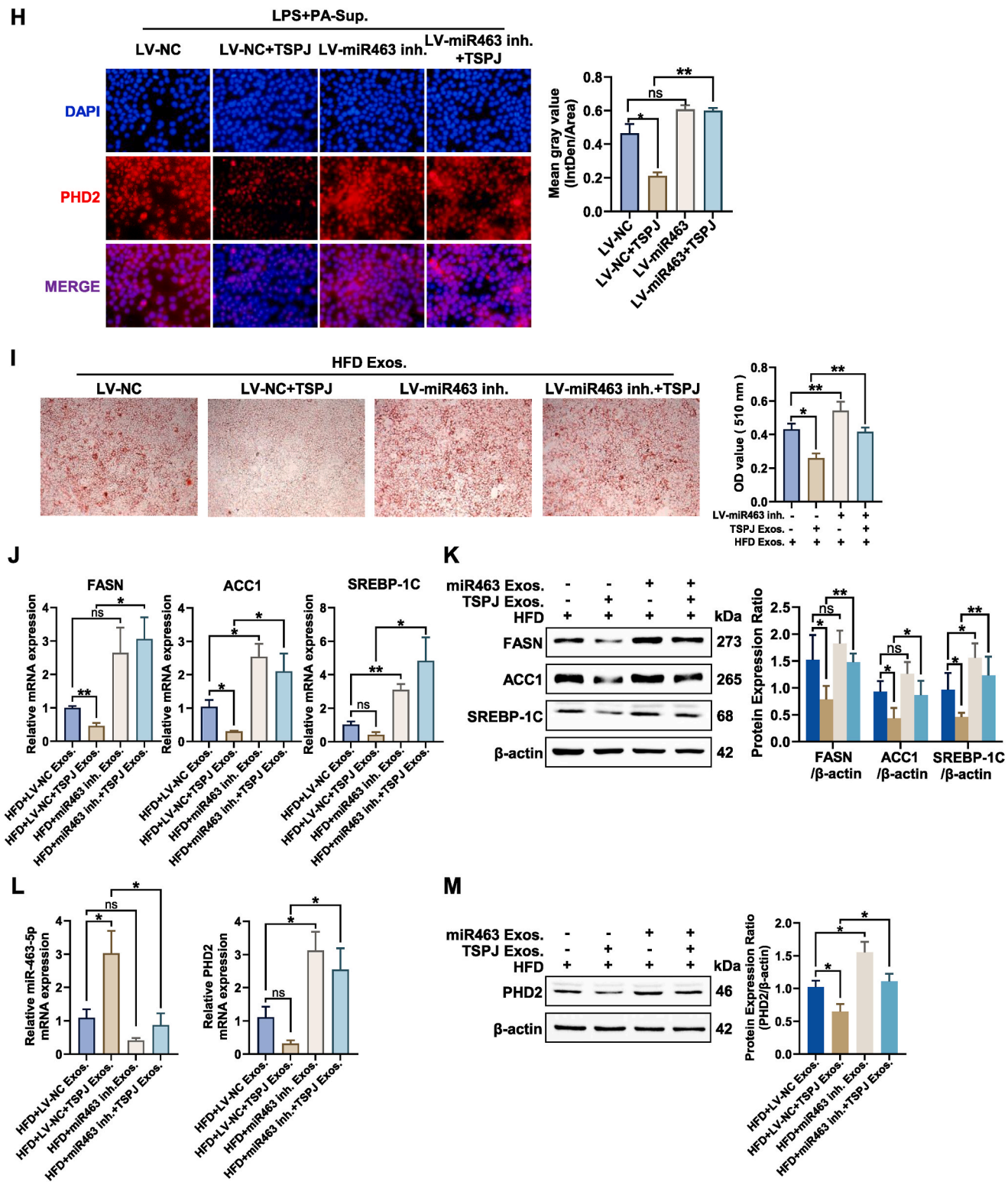


Fig. 8. (continued).

medicine *Panax japonicus* (T. Nees) C.A. Mey., have shown significant lipid-lowering and anti-inflammatory effects, particularly in alleviating chronic inflammation induced by HFD (Gao et al., 2021, 2023; Wang et al., 2021).

In this study, a lipotoxic inflammation cell model was created by inducing the RAW264.7 macrophage with LPS and PA, and a NASH animal model was created by giving HFD to C57BL/6J mouse. Both

models were treated with TSPJ and characterized extensively. For the HFD mice, the TSPJ treatment reduced the body weight and the liver weight, decreased the serum levels of TG, T-CHO, ALT, and AST, and weakened the accumulation of lipid droplets in the liver. The buildup of TG in hepatocytes is a significant contributor to liver inflammation (Wen et al., 2021). In addition, during the progression of NASH, macrophage polarization is often imbalanced. Classical M1 macrophages overexpress

markers such as CD80 and CD86, and they enhance local inflammatory responses by secreting large amounts of pro-inflammatory cytokines (TNF- α , IL-1 α , IL-6, IL-12), which further exacerbates liver damage. In contrast, activated M2 macrophages overexpress markers such as CD163, IL-10, and TGF- β , which exert anti-inflammatory effects and promoting tissue repair, and they are known to reduce systemic inflammation, remedy insulin resistance, and improve lipid metabolism (Yunna et al., 2020). In the liver of the HFD mice, the expression of M1 markers increased, and the expression of M2 markers declined. However, these adverse changes were reversed after TSPJ treatment, suggesting that TSPJ can suppress the M1 polarization and promote the M2 polarization of hepatic macrophages in the mouse model of NASH.

To further study the roles of macrophages, primary hepatic macrophages were isolated and extracted from the mouse models. Using both RAW264.7 cells and hepatic primary macrophages in the cell studies provided a more wholesome understanding of the underlying cell physiology. The cultures of both the primary hepatic macrophages and the RAW264.7 macrophages were cleansed by ultrafiltration using a 220 nm filter, and the resulting supernatant was used to culture AML12 hepatocytes. If the RAW264.7 macrophages were previously induced by LPS/PA, the AML12 cells exhibited significantly stronger expression of lipid synthesis-related genes and accumulation of lipid droplets. However, if the RAW264.7 macrophages were previously exposed to TSPJ, the AML12 cells exhibited decreased expression of lipid synthesis-related genes and had a lower number of lipid droplet vacuoles. Thus, the macrophages likely secreted factors into the supernatant of the culture that regulated lipid accumulation in hepatocytes. The exosomes derived from hepatic macrophages were therefore collected by ultracentrifugation and identified. The fluorescence staining analysis verified that the exosomes derived from the hepatic macrophages of HFD-induced mice could enter hepatocytes and mitigate lipid deposition. When the AML12 cells were incubated in the supernatant of the culture of the primary hepatic macrophages collected from mouse models, the lipid deposition in the AML12 cells was mitigated significantly if the mice were previously treated with TSPJ. However, when the AML12 hepatocytes were incubated in the supernatant of the macrophage culture that contained the exosome inhibitor GW4869, the expression of lipid synthesis-related genes in the AML12 cells were no longer reduced even if the macrophage was previously treated with TSPJ. Thus, TSPJ must have reduced hepatic lipid deposition through the exosomes derived from hepatic macrophages. The importance and specificity of liver macrophage-derived exosomes have been well documented, and they may be worthy targets in the development of new therapeutics (Fan et al., 2024; Shen et al., 2020; Xiang et al., 2023).

It is worth noting that miRNAs play a pivotal role in exosome-mediated crosstalk. Studies have shown that miR-210, highly expressed in the exosomes derived from high-glucose-induced macrophages, inhibits glucose uptake in adipocytes and mitochondrial respiratory chain activity, thereby contributing to insulin resistance and obesity (Tian et al., 2020). The expression of miR-463-5p in the liver tissues of the C57BL/6J mice was significantly decreased by HFD but significantly increased by TSPJ (Fig. 5A), and TSPJ partially offset the HFD-induced downregulation of miR-463-5p expression in the RAW 264.7 cells and the exosomes from the primary hepatic macrophages (Fig. 5B). According to bioinformatic analysis, *PHD2* was a potential downstream target of miR-463-5p. The specific binding between miR-463-5p and the 3' UTR of *PHD2* was confirmed by dual luciferase reporter assays. It has been reported that PHD2 is a key rate-limiting enzyme in the oxygen-sensing pathway. The impacts of TSPJ on PHD2 expression in the liver tissues of the mouse models and in the AML12 hepatocytes were evaluated using RT-qPCR, Western blot, and immunohistochemical staining. When the mice received TSPJ, the PHD2 expression in the liver tissue decreased significantly. However, when the exosome secretion inhibitor GW4869 was applied, the reduction in PHD2 expression in hepatocytes was minimal. Therefore, TSPJ regulated PHD2 expression in hepatocytes via exosomes. The relationship

between TSPJ, exosomal miR-463-5p, and hepatic PHD2 was further evaluated using miR-463-5p knockdown models. The miR-463-5p knockdown exacerbated lipid deposition and inflammatory responses in hepatocytes. Without miR-463-5p knockdown, TSPJ clearly elevated miR-463-5p expression and reduced PHD2 expression in hepatocytes. In the presence of miR-463-5p knockdown, TSPJ could not readily reduce the expression of lipid synthesis-related genes and M1 markers. That is, TSPJ upregulated the exosomal miR-463-5p of liver macrophages to inhibit PHD2 in hepatocytes.

Under aerobic conditions, PHD2 targets HIF-1 α via the von Hippel-Lindau protein (pVHL) for proteasomal degradation (Laitakari et al., 2019; Lee et al., 2016). When PHD2 is inhibited, HIF-1 α is not hydroxylated and thus escapes degradation. The stabilized HIF-1 α translocates to the nucleus, where it dimerizes with HIF-1 β to consequently activate the transcription of numerous genes involved in adaptive responses to low oxygen (Ju et al., 2016; Wu et al., 2018). Many research works have described the key role of HIF-1 α in improving liver health. For example, Yoo et al. reported that HIF-1 α expression protects HepG2 cells against fatty acid-induced toxicity (Yoo et al., 2014). Arai et al. found that the HIF-1 α -induced expression of lipin1 prevents abnormal lipid accumulation by inhibiting peroxisomal fatty acid oxidation (Arai et al., 2018). He et al. reported that silencing HIF-1 α exacerbates NAFLD *in vitro* by inhibiting PPAR- α /ANGPTL4 signaling (He et al., 2021). Fig. 9 draws the proposed mechanism of how TSPJ protected hepatocytes via exosomal miR-463-5p and hepatic PHD2.

There are currently no approved drugs for the clinical treatment of NASH. The exosomes derived from liver macrophages carry selectively secreted proteins and miRNAs in response to NASH-related signals, and they provide a potential approach for targeted therapy. Enhancing the delivery of beneficial miRNAs to liver cells is expected to correct hepatic inflammation and metabolic dysregulation. Given the complex pathophysiology of NASH, treatment requires the involvement of multiple targets and pathways. Unlike single-target therapies, the natural product TSPJ, with multiple pharmacological actions, may offer a more comprehensive intervention in the complex pathological processes of NASH. This is especially beneficial in the early stages of the disease, where TSPJ can improve lipid metabolism, reduce inflammation, and prevent liver damage. Nevertheless, our study still has certain limitations. While we have found that TSPJ inhibits the progression of NASH through the miR-463-5p/PHD2 axis, the role of downstream genes and pathways of PHD2 has not been thoroughly explored, especially the protective and harmful roles of HIF-1 α in acute and chronic liver diseases, and further investigation is needed (Ju et al., 2016). In addition, since the HFD-induced mice only received a short-term intervention of four months, the long-term effects of TSPJ may vary and need to be tested in future studies.

5. Conclusions

This study revealed that TSPJ, the main active component of *Panax japonicus* (T. Nees) C.A. Mey., possessed anti-inflammatory, lipid-lowering, and liver-protective effects. We demonstrated that TSPJ induced M2 polarization in hepatic macrophages and promoted the expression of miR-463-5p in macrophage-derived exosomes, which alleviated the hepatic inflammation induced by HFD and reduced the hepatic lipid accumulation since the upregulated miR-463-5p inhibited PHD2 in hepatocytes. By modulating the crosstalk between hepatic macrophages and hepatocytes, TSPJ mitigated hepatic lipid metabolism disorders in NASH models. These findings provide fresh perspectives for future research and the development of drugs in preventing and treating NAFLD.

CRedit authorship contribution statement

Jingjie Yang: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Yibei Gui:** Formal analysis,

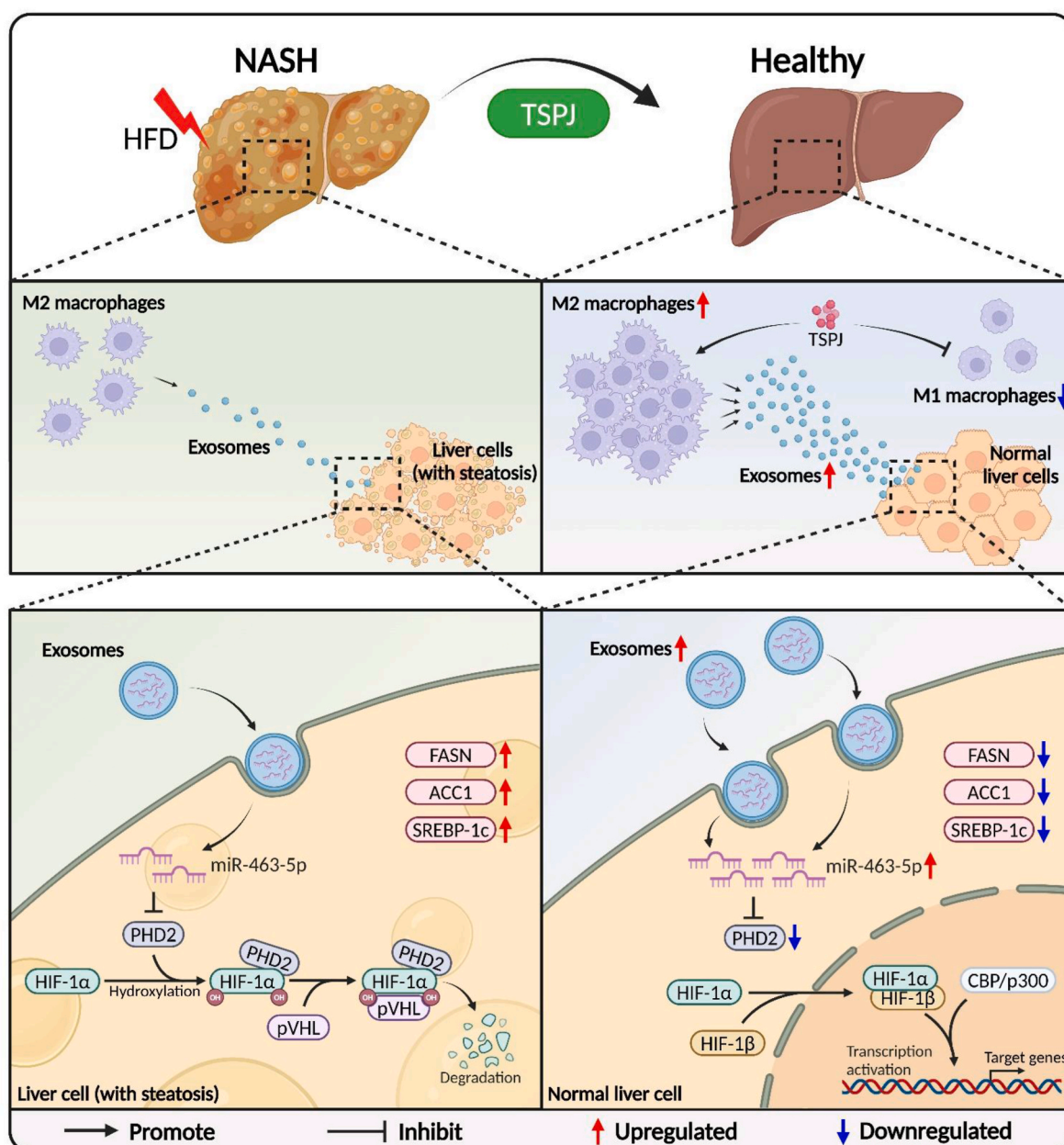


Fig. 9. Mechanistic illustration of the reduction of hepatic lipid accumulation and inflammatory response by TSPJ. TSPJ promoted the M2 polarization of macrophage and increased the level of miR-463-5p in exosomes, which reduced PHD2 expression in hepatocytes and affected HIF-1 α degradation and activation, ultimately improved lipid metabolism. CBP, cyclic adenosine monophosphate response element-binding protein; pVHL, von Hippel Lindau protein; p300, coactivator acetyltransferase. The artwork was created using BioRender.

Investigation, Methodology, Visualization, Writing – review & editing. **Ying Zheng:** Writing – original draft, Methodology, Formal analysis, Data curation. **Haodong He:** Writing – original draft, Formal analysis, Data curation. **Lihan Chen:** Visualization, Software. **Tongtong Li:** Software, Investigation. **Haoran Liu:** Data curation. **Dongshuo Wang:** Investigation. **Ding Yuan:** Validation, Supervision, Resources, Conceptualization. **Chengfu Yuan:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition.

Data availability statement

All data supporting this study's findings are available from the corresponding author upon reasonable request unless they cannot be shared due to privacy or ethical restrictions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2025.119376>.

Data availability

Data will be made available on request.

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